CS 1675 · GROUP PROJECT — MILESTONE 2

What Does the Model Actually Read?

Machine Learning Based Fake News Detection · progress report on data, dual-encode baselines, and a shortcut diagnosis

GROUP MEMBERS

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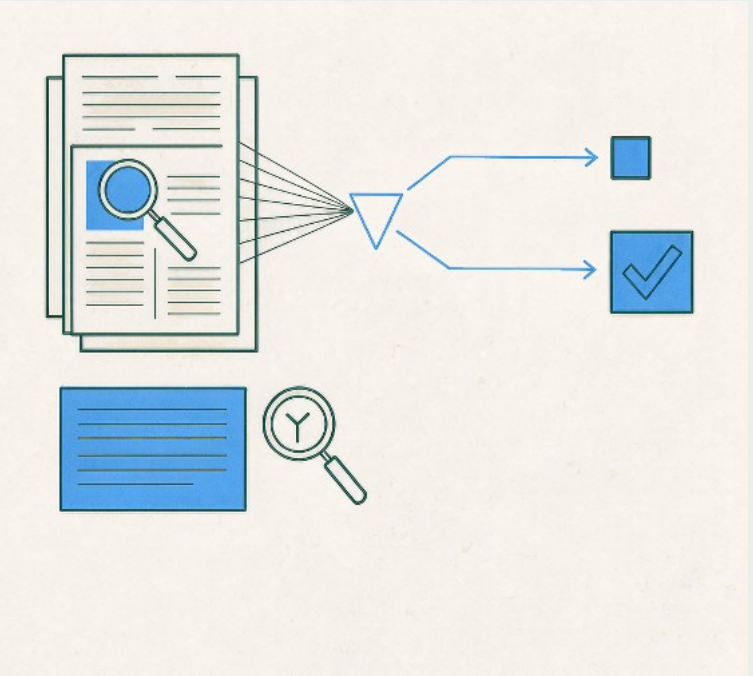
王浩懿 2024141520231

DATE

October 27 2026

REPO

github.com/Dande-lifeon/fake-news-detection



Data: 44,898 → 38,503, With a Label Caveat



Cleaning is label-asymmetric: −26.3% of fake rows vs −1.0% of real rows

fake share shifts 52.3% → 45.0% · subject & date: audit-only, never features

38,503 = 21,194 real / 17,309 fake

CLEANING FUNNEL · ALL NUMBERS FROM pd.read_csv

Cleaning is label-asymmetric: **−26.3% of fake vs −1.0% of real** → fake share shifts 52.3% → 45.0%.

SPLITS

26,951 / 3,851 / 7,701

stratified · seed 42 · **test read once**

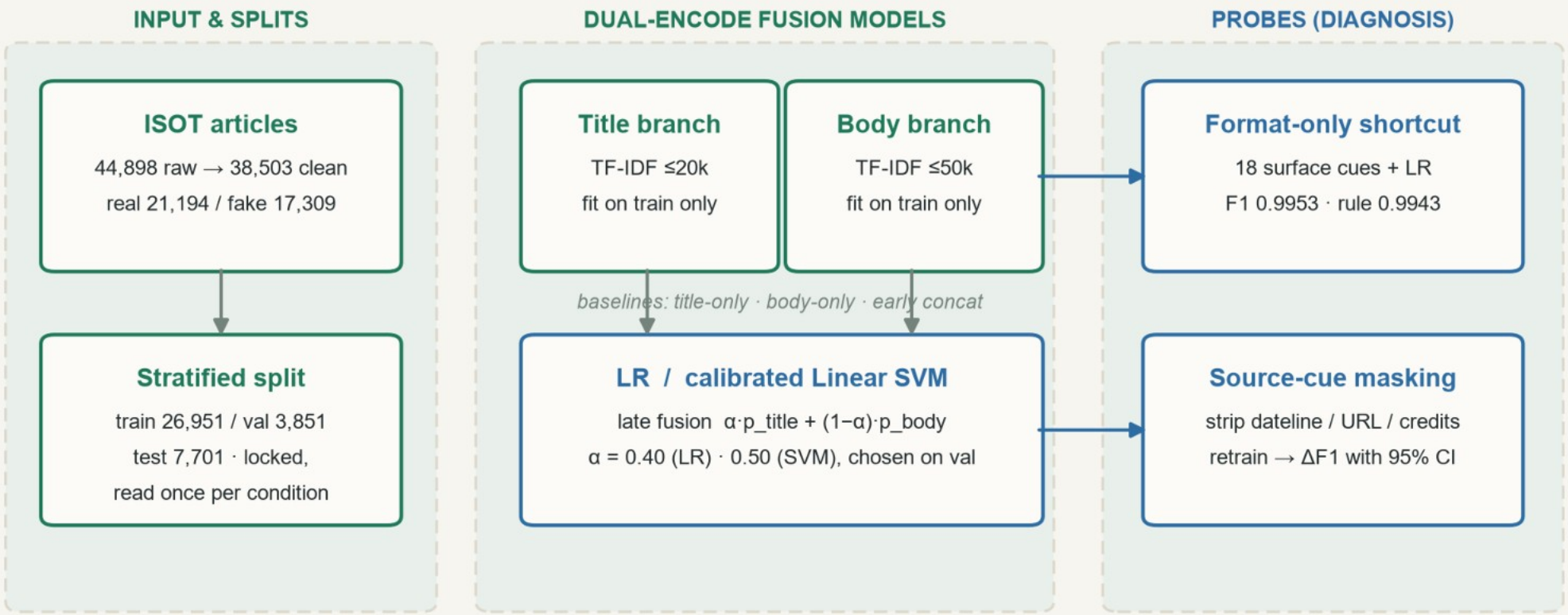
LABEL

1 = fake — by source (Reuters vs flagged sites), not per-article fact-checking.

AUDIT-ONLY

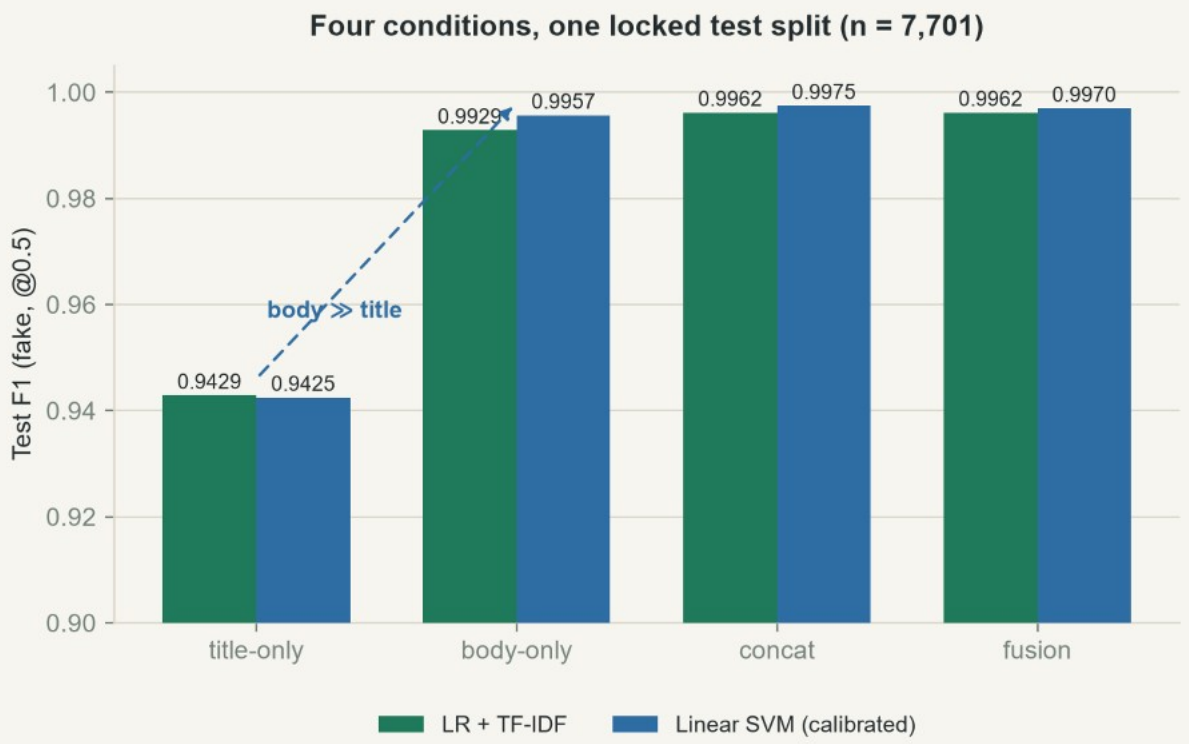
subject & date never enter any feature — they would leak the label directly.

Models: Dual-Encode Fusion · 4 Conditions



vectorizers fit on train only · α & C chosen on val · test evaluated once · fake = positive class

Results: F1 0.99+ — the Ablation Story



TEST F1 @0.5 · n = 7,701 · FAKE = POSITIVE

MODEL	TITLE	BODY	CONCAT	FUSION	α
LR + TF-IDF	0.9429	0.9929	0.9962	0.9962	0.40
Linear SVM	0.9425	0.9957	0.9975	**0.9970**	0.50

fusion − concat ΔF1 ≤ 0.0006 · McNemar p = 0.39–0.80

01 BODY ≫ TITLE

Titles alone are weak (0.94); bodies carry the signal.

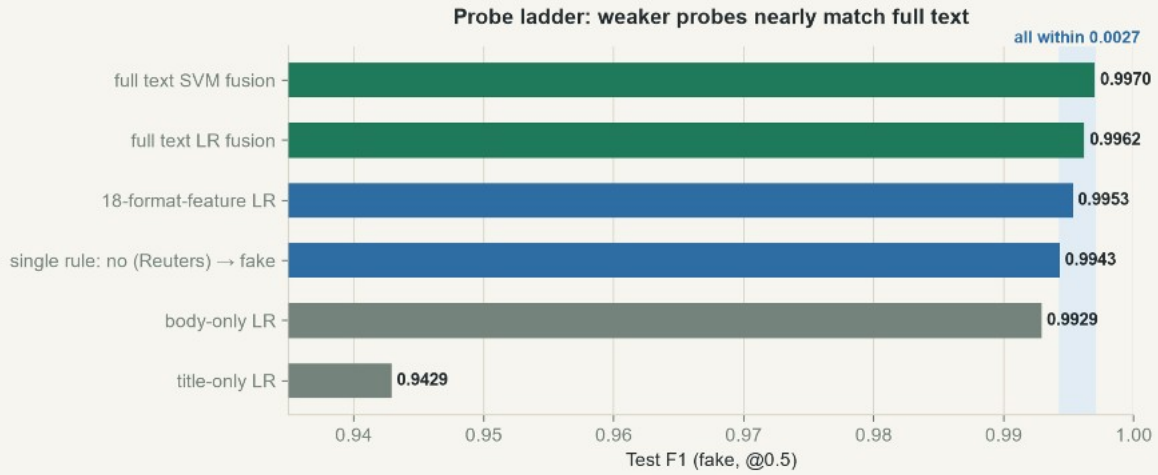
02 CONCAT ≈ FUSION

Late fusion adds nothing significant over simple concatenation.

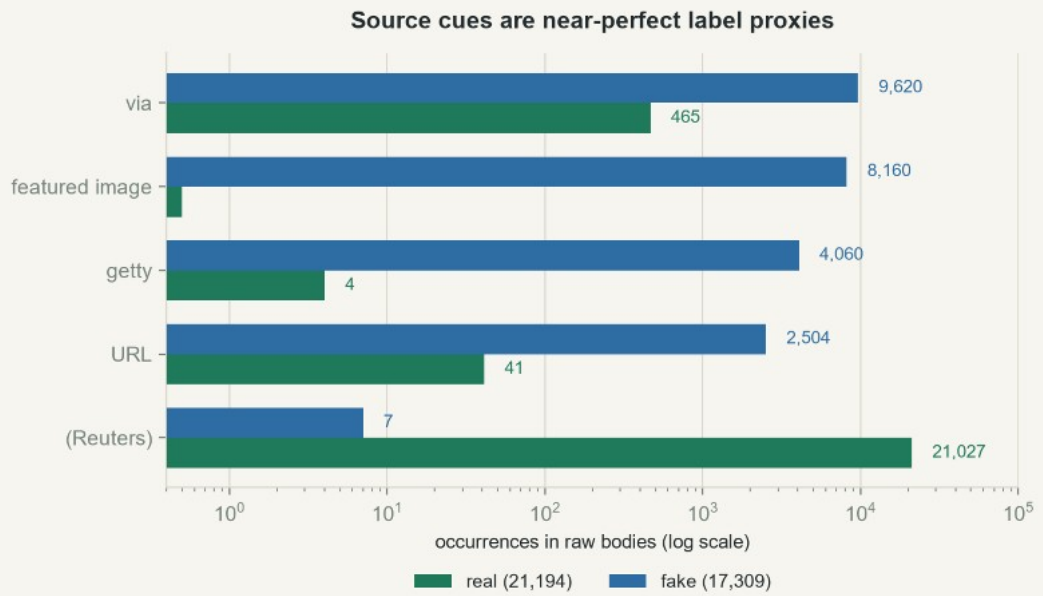
03 LR ≈ SVM

Model choice does not matter on ISOT — gaps are noise.

The Shortcut: You Don’t Need to Read the News



PROBE LADDER · WEAKER PROBES NEARLY MATCH FULL TEXT



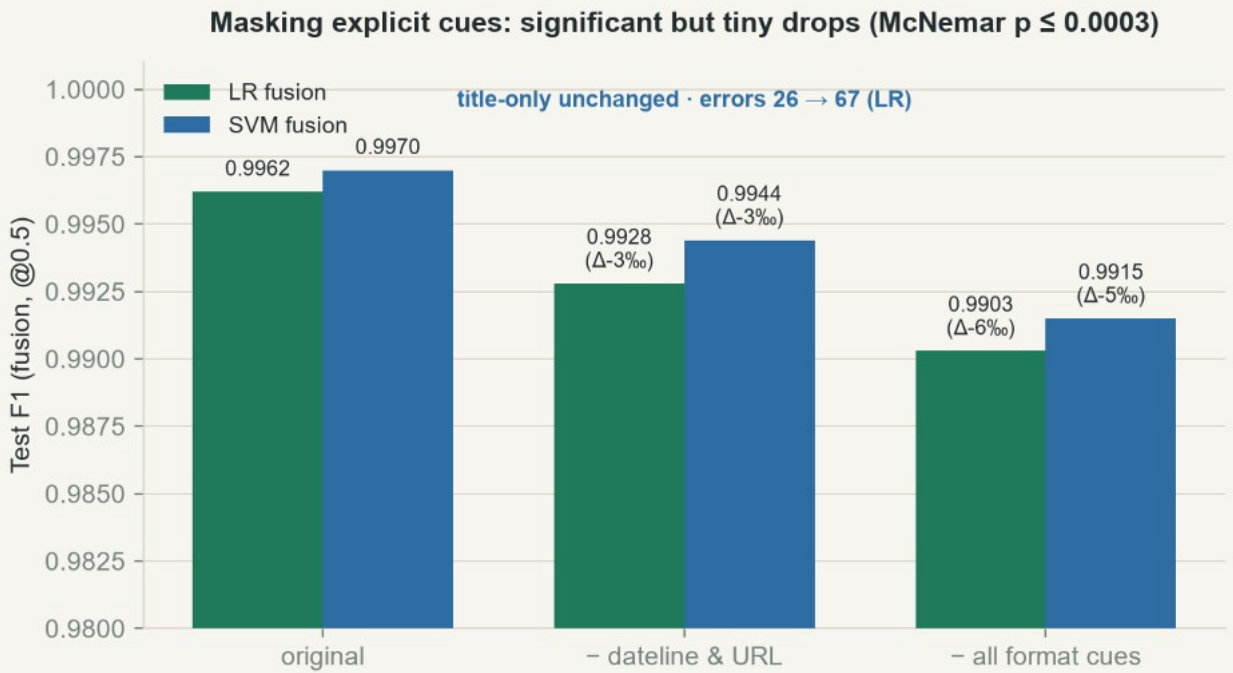
SOURCE-CUE HITS IN RAW BODIES · LOG SCALE

REAL · TRUE.CSV *“WASHINGTON (Reuters) - The head of a conservative Republican faction…”*

FAKE · FAKE.CSV *“…Featured image via David McNew/Stringer/Getty Images”*

18 format features alone: F1 0.9953 — statistically indistinguishable from full text ($p = 0.13–0.43$).

Problems: Masking Helps — But Not Enough



FUSION TEST F1 UNDER SOURCE-CUE MASKING · Δ IN PERMILLE

ΔF1 = −0.0026…−0.0059 · McNemar p ≤ 0.0003 · title-only unchanged · errors 26 → 67 (LR).

01 RESIDUAL IS STILL 0.99+

The shortcut is distributed — topics, headline style, CMS traces — so single-point masking cannot remove it.

02 NO STATISTICAL HEADROOM

Model gaps (≤ 0.001) sit far below seed noise; we currently have a single seed per configuration.

03 BENCHMARK LIMITS

1.3% near-duplicates are not group-split yet; labels are source-based, not fact-checked.

Next Plan: From High Scores to Trustworthy Scores

NEURAL MODEL

Bi-LSTM & DistilBERT dual-encode under the same protocol (free Colab T4)

ADDITIONAL

Probe ladder & balanced-source test; group split for the 1.3% near-duplicates

MORE TEST

3 seeds, mean ± std · cross-dataset (LIAR) zero-shot · final 2-page report

*Goal: report **what the model reads**, not just how well it scores.*

High scores ≠ understanding —

we measure what the model reads.

THANK YOU AND QUESTIONS

repo · github.com/Dande-lifeon/fake-news-detection · 1-page report submitted